‘MAGNETIC FIELDS & FORCES:
FROM CURRENTS & MOVING CHARGES EXPLAINED’

Scoolam24

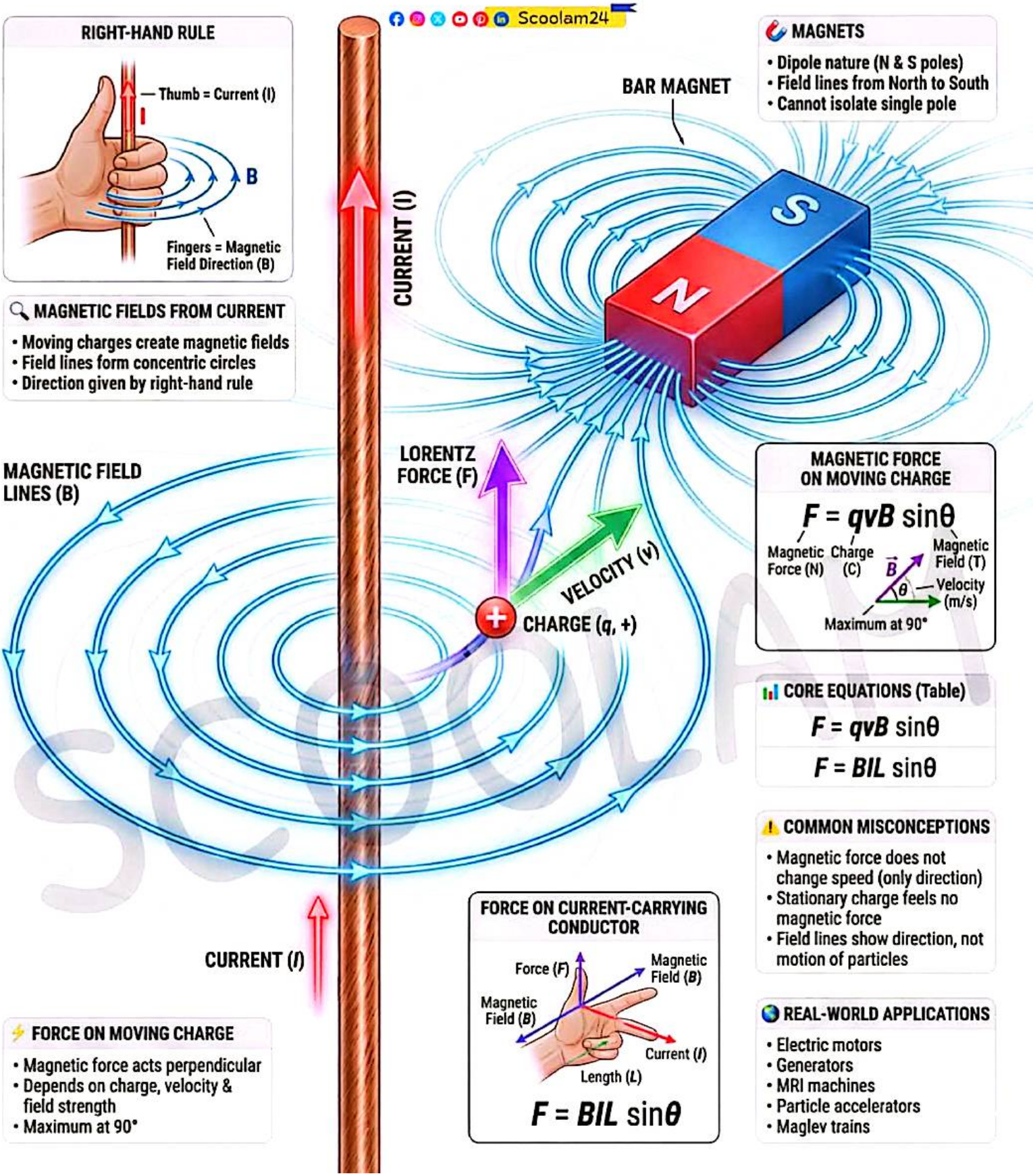
RIGHT-HAND RULE

MAGNETIC FIELDS FROM CURRENT

- Moving charges create magnetic fields
- Field lines form concentric circles
- Direction given by right-hand rule

MAGNETS

- Dipole nature (N & S poles)
- Field lines from North to South
- Cannot isolate single pole



MAGNETIC FORCE ON MOVING CHARGE

$$F = qvB \sin\theta$$

Magnetic Force (N) — Charge (C) — Magnetic Field (T) — Velocity (m/s)

Maximum at 90°

CORE EQUATIONS (Table)

$$F = qvB \sin\theta$$

$$F = BIL \sin\theta$$

COMMON MISCONCEPTIONS

- Magnetic force does not change speed (only direction)
- Stationary charge feels no magnetic force
- Field lines show direction, not motion of particles

FORCE ON CURRENT-CARRYING CONDUCTOR

Force (F) — Magnetic Field (B) — Magnetic Field (B) — Current (I) — Length (L)

$$F = BIL \sin\theta$$

FORCE ON MOVING CHARGE

- Magnetic force acts perpendicular
- Depends on charge, velocity & field strength
- Maximum at 90°

REAL-WORLD APPLICATIONS

- Electric motors
- Generators
- MRI machines
- Particle accelerators
- Maglev trains

الممسوحة ضوئيا بـ CamScanner

'FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION & LENZ'S LAW EXPLAINED

Scoolam24

WHAT IS ELECTROMAGNETIC INDUCTION?

- Changing magnetic field induces EMF
- Discovered by Faraday
- Foundation of generators

FARADAY'S LAW EQUATION INSET.

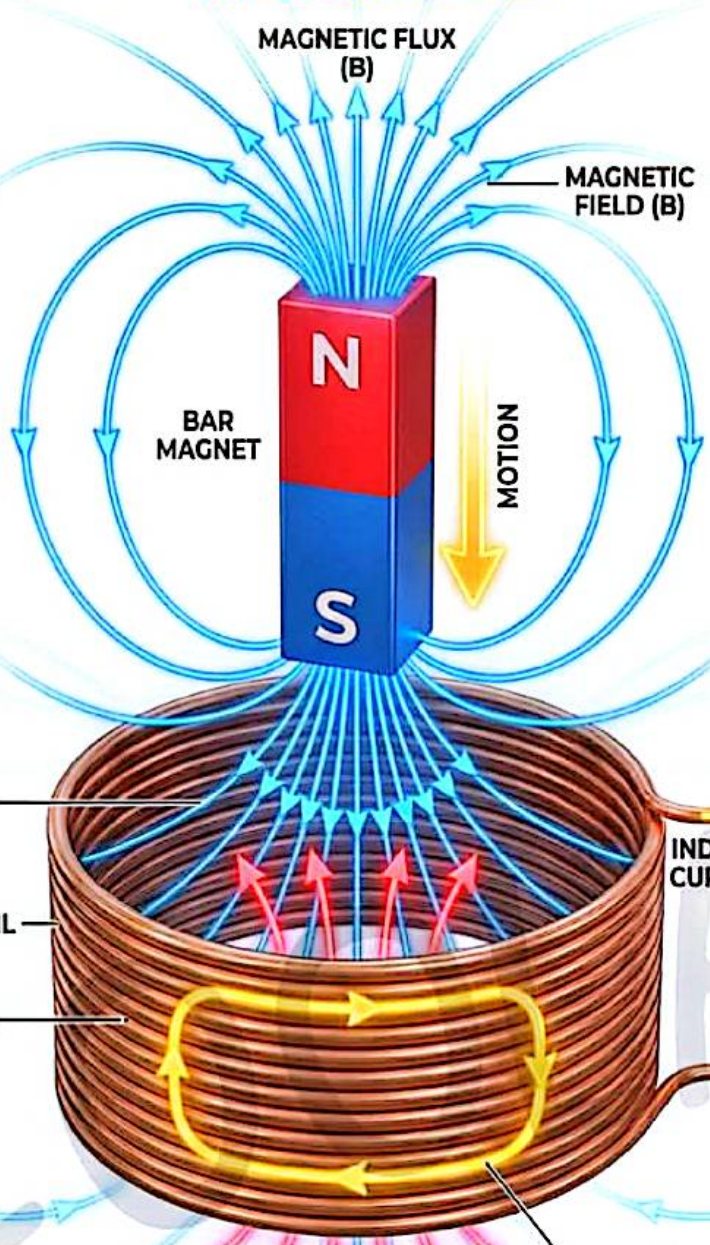
FARADAY'S LAW EQUATION

CHANGE IN FLUX

$$E = -\frac{d\Phi_B}{dt}$$

INDUCED EMF — TIME INTERVAL

- Faster change → larger EMF
- Negative sign → Lenz's Law



MAGNETIC FLUX INSET.

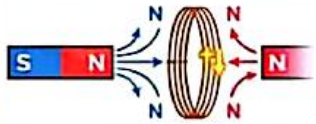
MAGNETIC FLUX (Φ_B)

$$\Phi_B = B \cdot A \cdot \cos\theta$$

- Depends on field strength
- Depends on area
- Depends on orientation

LENZ'S LAW EXPLANATION INSET.

LENZ'S LAW EXPLANATION



- Induced current opposes change
- Conservation of energy principle
- FIELD INTERACTION DIAGRAM: Induced N opposes approaching N

GALVANOMETER

COMMON MISCONCEPTIONS

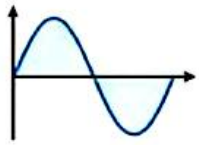
- Strong magnet alone doesn't induce current
- Motion is not required if flux changes
- Direction matters

REAL-WORLD APPLICATIONS

- Power plants
- Electric generators
- Transformers
- Induction cooktops
- Wireless charging

AC GENERATOR SCHEMATIC INSET.

AC GENERATOR SCHEMATIC



OUTPUT WAVEFORM (EMF vs. Time)

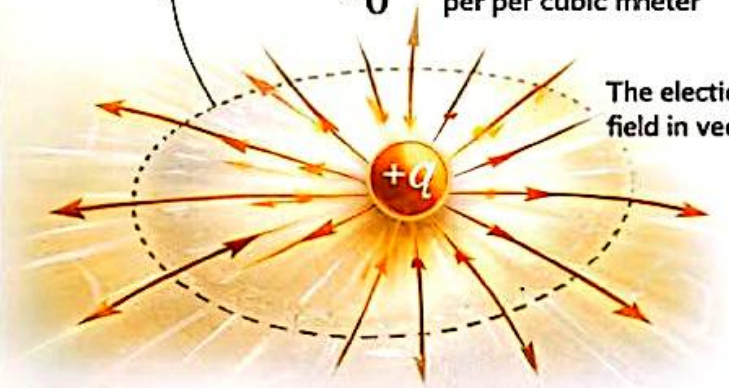
الممسوحة ضوئيا بـ CamScanner

Gauss's Law for Electricity

$$\vec{\nabla} \circ \vec{E} = \frac{\rho}{\varepsilon_0}$$

The electric field in coulombs per per cubic mneter

The electic field in vector



The dot product turns the del operator into the divergence

Faraday's Law of Induction

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

The rate of change of the magnetic field with time

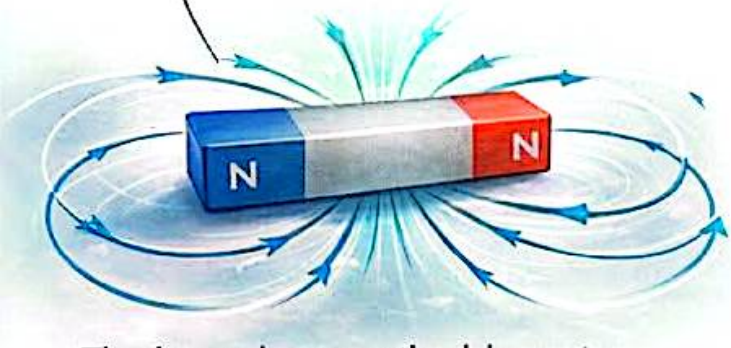
The dot product turns the del operator

The rate of change of the magnetic field with time

Gauss's Law for Magnetism

$$\vec{\nabla} \circ \vec{B} = 0$$

The magnetic field in Teslas

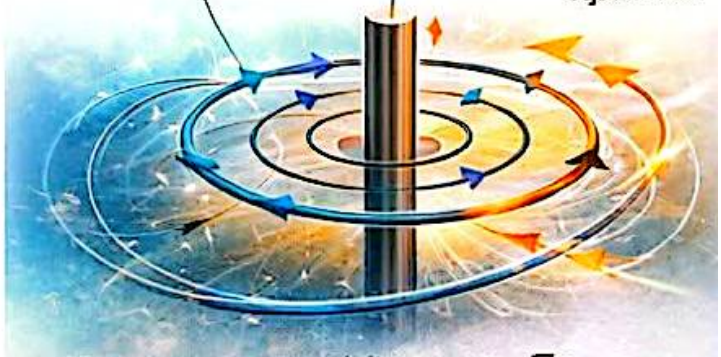


The dot product turns the del operator into the divergence

Ampére's Law with Maxwell's Correction

$$\vec{\nabla} \times \vec{B} = \mu_0 J + \frac{\partial \vec{E}}{\partial t}$$

The electric current density in amperes per sejubis meter



The magnetic permeability of free space $m_p^{\cdot} - E_{\bar{o}}$

Gauss's Law for Magnetism

$$\vec{\nabla} \circ \vec{B} = 0$$

The magnetic field in Teslas

The dot product turns the del operator into the divergence

Ampére's Law with Maxwell's Correction

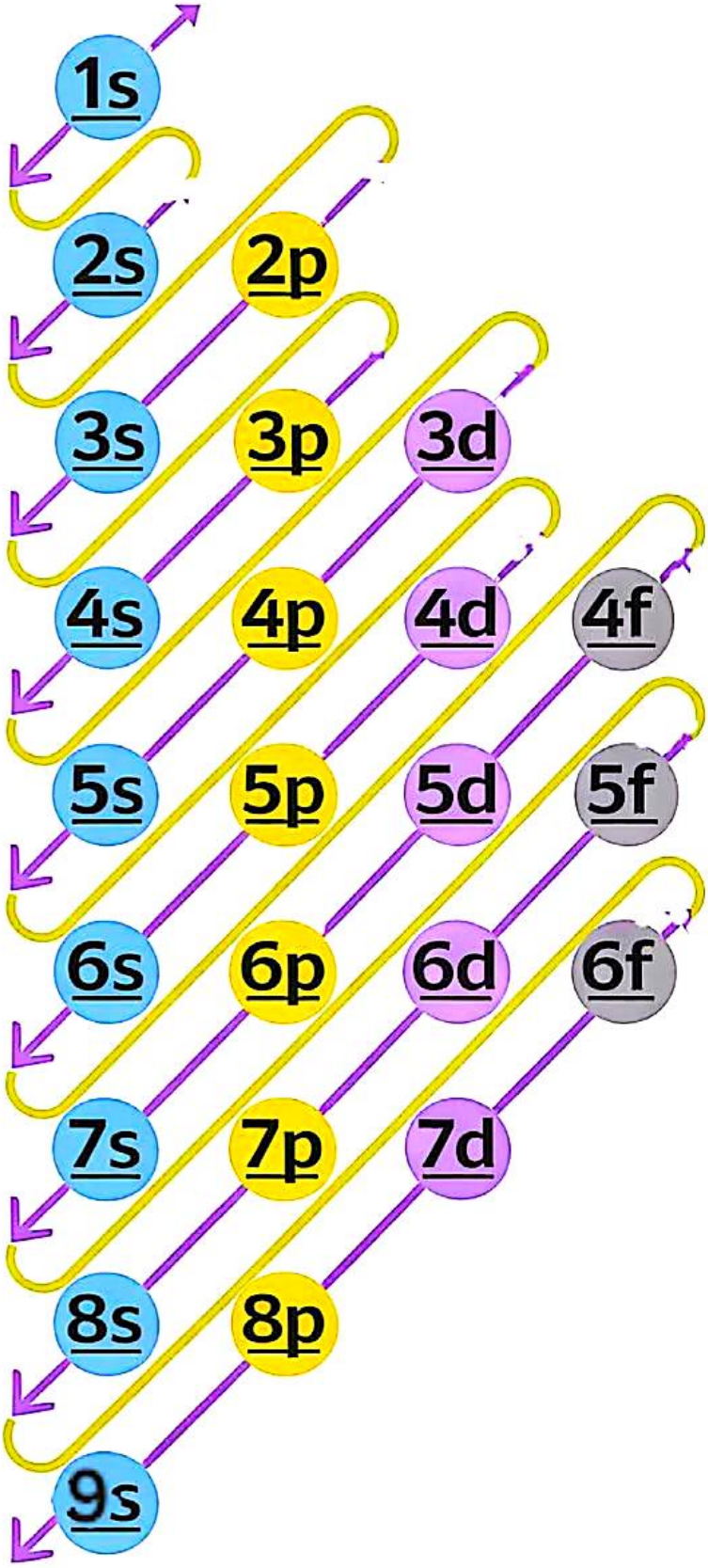
$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \frac{\partial \vec{E}}{\partial t}$$

The magnetic permeability of free space

the electric current density in amperes per square meter

الممسوحة ضوئيا بـ CamScanner

Aufbau Principle



s = 2 p = 6 d = 10 f = 14

الممسوحة ضوئيا بـ CamScanner